

REAL TIME INSTANT WEATHER PREDICTOR

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OUTLINE

1. Abstract

2. Existing System

3. Proposed System

4. System Architecture

5. Project Modules

6. Objectives

7. CO'S Attainments-FWD

1. CO1

2. CO2.....5.CO5

8.CO'S Attainments-DSA

1. CO1

2. CO2.....5.CO5

9.Screen shorts for FWD & DSA

10.Conclusion

11.Gio-Tagged photos

ABSTRACT



SkyCast is a weather prediction web application built using HTML5, CSS3, and JavaScript that works completely offline without any server, API, or framework. The project is created as a single .html file that opens directly in any browser including VS Code. The app allows users to search weather information for cities around the world and displays current weather details like temperature, humidity, wind speed, and pressure. It also shows a 7-day weather forecast with actual day names like Monday and Tuesday, correct weather symbols like sun for sunny, cloud for cloudy, and rain for rainy days, along with temperature ranges and rainfall amounts.

The app is designed in three parts — HTML builds the structure of the page, CSS handles the animations and visual design including four background scenes that change based on the city's local time such as daytime, sunrise, sunset, and nighttime with stars and moon, and JavaScript handles all the logic like searching cities, switching between Celsius and Fahrenheit, saving recent searches, and displaying weather data on screen. Each weather condition is identified using a number code called `wid`, and a function called `getIcon` reads that number and shows the correct weather symbol automatically. This project proves that a fully working, good-looking weather application can be built using only basic web technologies without needing any backend or paid API service.

Existing System

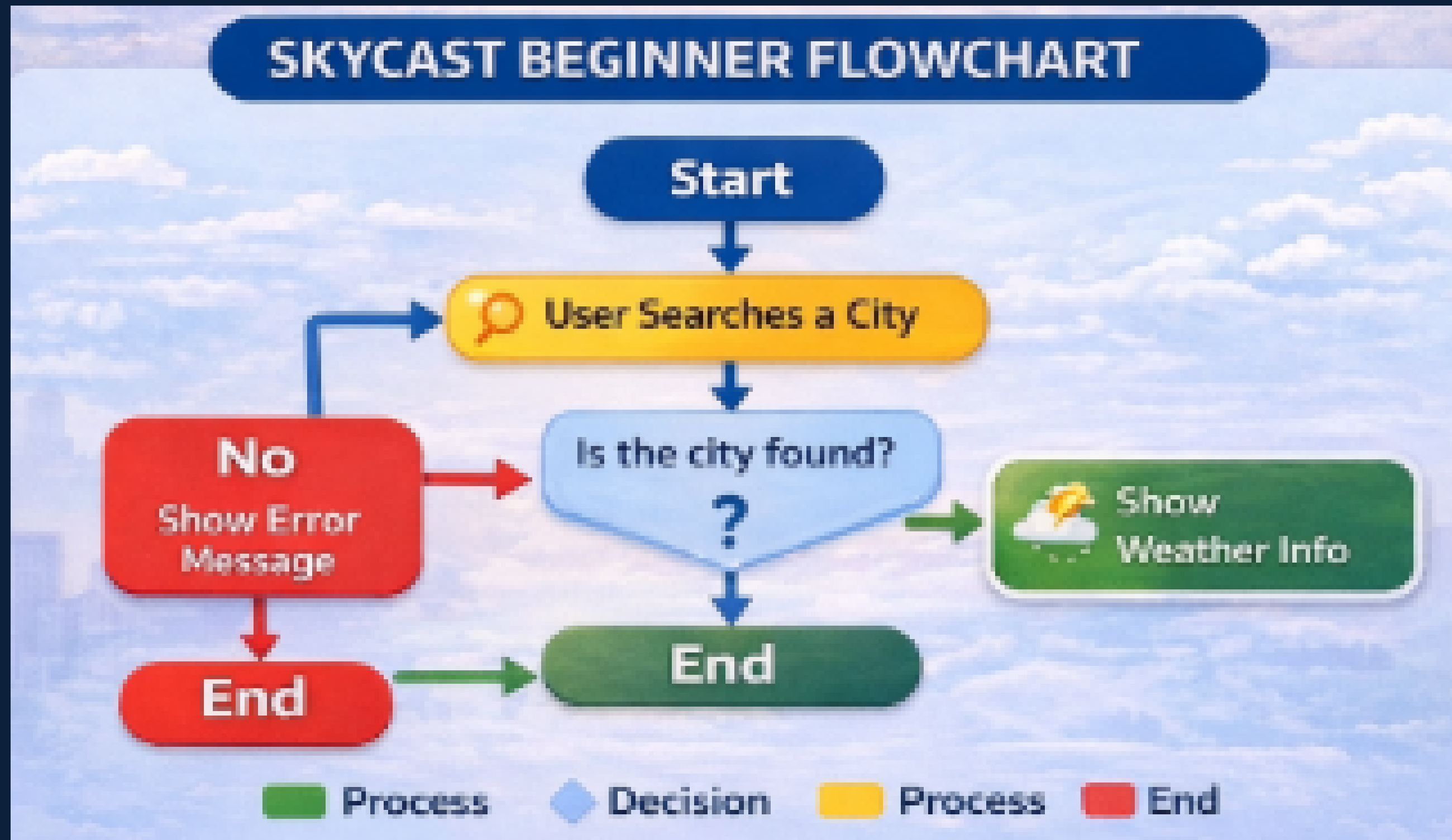
Popular weather platforms like AccuWeather, Weather.com, and OpenWeatherMap:

- Require live internet connection and paid API keys to function
- Depend on backend servers — not suitable for offline or frontend-only use
- Complex setup with authentication, rate limits, and third-party dependencies
- No single-file deployment — require Node.js, databases, or cloud hosting
- Static weather widgets offer no interactivity — no search, no unit toggle, no animations
- Existing templates show hardcoded data with no DOM manipulation or localStorage
- Not designed for academic/educational demonstration of web development concepts
- No animated backgrounds tied to real-time day/night detection
- Difficult to customize UI to match specific design or course requirements

Proposed System — SkyCast

- Single .html file — no backend, no server
- No API key — 12 cities embedded in JavaScript
- Works offline — opens in any browser instantly
- Animated scenes — Day / Dawn / Dusk / Night
- 7-day forecast with temperature range bars
- Fuzzy search + °C/°F toggle
- localStorage — saves recent searches
- Responsive — mobile & desktop ready
- Covers CO1–CO5 — HTML, CSS, JS, DOM, ES6
- No frameworks — pure Vanilla JavaScript

FLOW CHART



System Architecture



What is inside the single weather-app.html file?

📄 HTML — The Skeleton

- Search Box
Where you type the city name
- Weather Card
Shows city, temperature, icon
- Stats Row
Humidity, Wind, Pressure, Visibility
- Sun and Sky Card
Sunrise, Sunset, Cloud cover
- Forecast Table
7 days of weather in a list
- Footer
Project info at the bottom

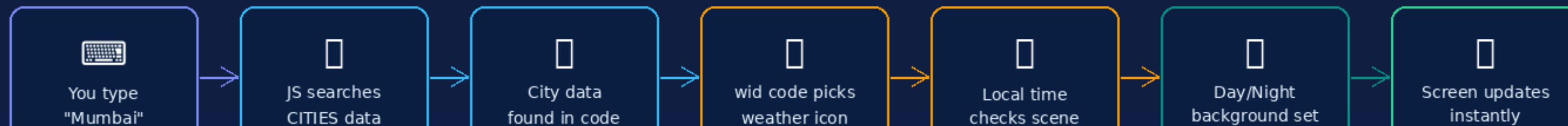
📄 CSS — The Look

- Dark Navy Theme
Background colour and text colour
- Animated Sun
Spinning rays and glowing effect
- Moving Clouds
Drift across the sky slowly
- Twinkling Stars
Appear at night and blink
- Responsive Design
Works on mobile and desktop
- Glass Cards
Frosted glass effect on boxes

⚙️ JavaScript — The Brain

- City Search
Finds the city you typed
- Weather Data
12 cities stored inside JS code
- Forecast Days
Shows Monday, Tuesday etc.
- Weather Icons
Sun, Rain, Snow, Thunder icons
- Time Detection
Shows Day / Night background
- Save History
Remembers your last searches

Step by Step — What happens when you search a city?



📄 Normal Apps Need:

Server + Database + API Key + Internet + Backend code

📄 SkyCast Only Needs:

One .html file + A web browser = Fully working weather app!

Project Overview

What is SkyCast?

SkyCast is a single-file weather predictor built entirely with HTML, CSS, and JavaScript.

No backend. No API key. No framework.

It demonstrates all 5 course outcomes (CO1-CO5) through a real, interactive, production-quality web application.

12

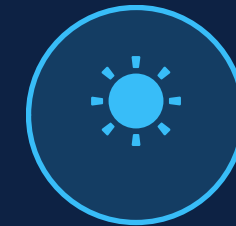
Cities

7

Day Forecast

4

Scenes



Animated Backgrounds

Day, Dawn, Dusk & Night scenes that respond to each city's local time



7-Day Forecast

Full weekly forecast per city — horizontal table with temp bars



Fully Offline

Zero internet required for weather data — everything runs in the browser

Key Features



Smart City Search

Fuzzy matching across 12 cities. Finds Tokyo even if you type 'toky'.



Live °C / °F Toggle

All temperatures recalculate instantly — today and all 7 forecast days.



Time-Based Scenes

Animated day, dawn, dusk and night backgrounds based on local city time.



Temperature Bar Chart

Proportional gradient bars show each day's high/low range at a glance.



Recent City History

localStorage saves your last 5 searches as clickable chips.



100% Offline

All weather data is embedded. Works with no internet after first load.

Course Outcomes Covered FWD

CO1	CO2	CO3	CO4	CO5
HTML & CSS Basics • Semantic HTML5 structure • External Google Fonts (HTTPS) • CSS variables & reset • Typography & color system	Advanced CSS • Flexbox & Grid layouts • CSS keyframe animations • Media queries / responsive • Custom properties (themes)	JavaScript Core • Variables, loops, functions • Objects & arrays • Arrow functions • Array methods (map/filter)	DOM & Events • getElementById / innerHTML • Event listeners & forms • localStorage (recent cities) • setTimeout (async UX)	ES6 & Best Practices • try/catch error handling • Form validation & regex • SEO meta tags • Modular code structure

COURSE OUT COMES COVERED IN DSA

C01	C02	C03	C04	C05	C06
Searching & Sorting	Linked List ADT	Stack, Queue & Heap	HashMap	Practical App	Menu-Driven Program
- Linear Search $O(n)$ - Binary Search $O(\log n)$ - Bubble / Selection / Insertion Sort - Merge Sort / Quick Sort 100K run benchmark timing	- Singly Linked List - insertFront $O(1)$ - removeLast $O(n)$ - traverse $O(n)$ - Duplicate removal	- Array-based Stack push/pop $O(1)$ - Circular Queue enqueue/dequeue $O(1)$ - Max-Heap Priority Queue $O(\log n)$ - Undo history (Stack) - Alert ticker (Circular Q)	- HashMap<String, City> $O(1)$ - Fuzzy fallback search $O(n)$ - Collections.sort() keys 12-city direct lookup - Collision handling	- All C01-C04 combined 9-option console menu - Scanner input handling - ANSI formatted output - Real-world integration	- Search city weather - Recent searches (LinkedList) - Sort & search demo - Heat ranking (PriorityQueue) - HashMap city lookup

SCREEN SHORT FOR FWD



SCREENSHOTS FOR DSA

```
C:\java program>java weatherapp.java
```

```
=====
SKYCAST - Instant Weather Predictor
Java Console | Data Structures Demo
C01 C02 C03 C04 C05 C06
=====
```

SKYCAST MENU

- 1. Search City Weather
- 2. Show Recent Searches (C02 - Linked List)
- 3. Show Search History (C03 - Stack)
- 4. Show Weather Alerts (C03 - Circular Queue)
- 5. Show Heat Ranking (C03 - Priority Queue)
- 6. Show Sorting + Search (C01)
- 7. Show HashMap Info (C04)
- 8. Toggle Celsius/Fahrenheit
- 9. List All Cities
- 0. Exit

```
=====
Enter choice: 1
```

```
Enter city name: hyderabad
```

SKYCAST - Hyderabad, IN

```
=====
Location   : Hyderabad, IN | Lat: 17.38 Lon: 78.49
Condition  : [FewCld ] Mostly Sunny
Temperature : 28 C
Feels Like  : 27 C
=====
```

WEATHER STATS

```
=====
Humidity    : 58 %
Wind Speed   : 3.6 m/s
Pressure     : 1013 hPa
Visibility   : 9 km
=====
```

SUN & SKY

```
=====
Sunrise     : 04:30
Sunset      : 16:00
Cloud Cover  : 20 %
=====
```

TEMPERATURE RANGE

```
=====
Max Temp    : 31 C
Min Temp    : 22 C
=====
```

7-DAY WEATHER FORECAST

Day	Condition	Description	High	Low	Rain
Today	[FewCld]	Mostly Sunny	31 C	21 C	--
Tomorrow	[Sunny]	Sunny	33 C	22 C	--
Wednesday	[PrtCld]	Partly Cloudy	30 C	21 C	--
Thursday	[Rain]	Light Rain	27 C	20 C	6 mm
Friday	[Cloudy]	Overcast	26 C	20 C	3 mm
Saturday	[PrtCld]	Partly Cloudy	29 C	21 C	--
Sunday	[Sunny]	Clear Sky	32 C	22 C	--

```
=====
```

SUMMARY

What We Built

SkyCast proves that a production-quality weather app needs no backend, no framework, and no API key.

- CO1-CO5 all demonstrated in one file
- Animated scenes (day/dawn/dusk/night)
- 7-day forecast with temperature bars
- °C / °F toggle with instant recalc
- Recent searches via localStorage

1

File

Everything in one .html

0

Errors

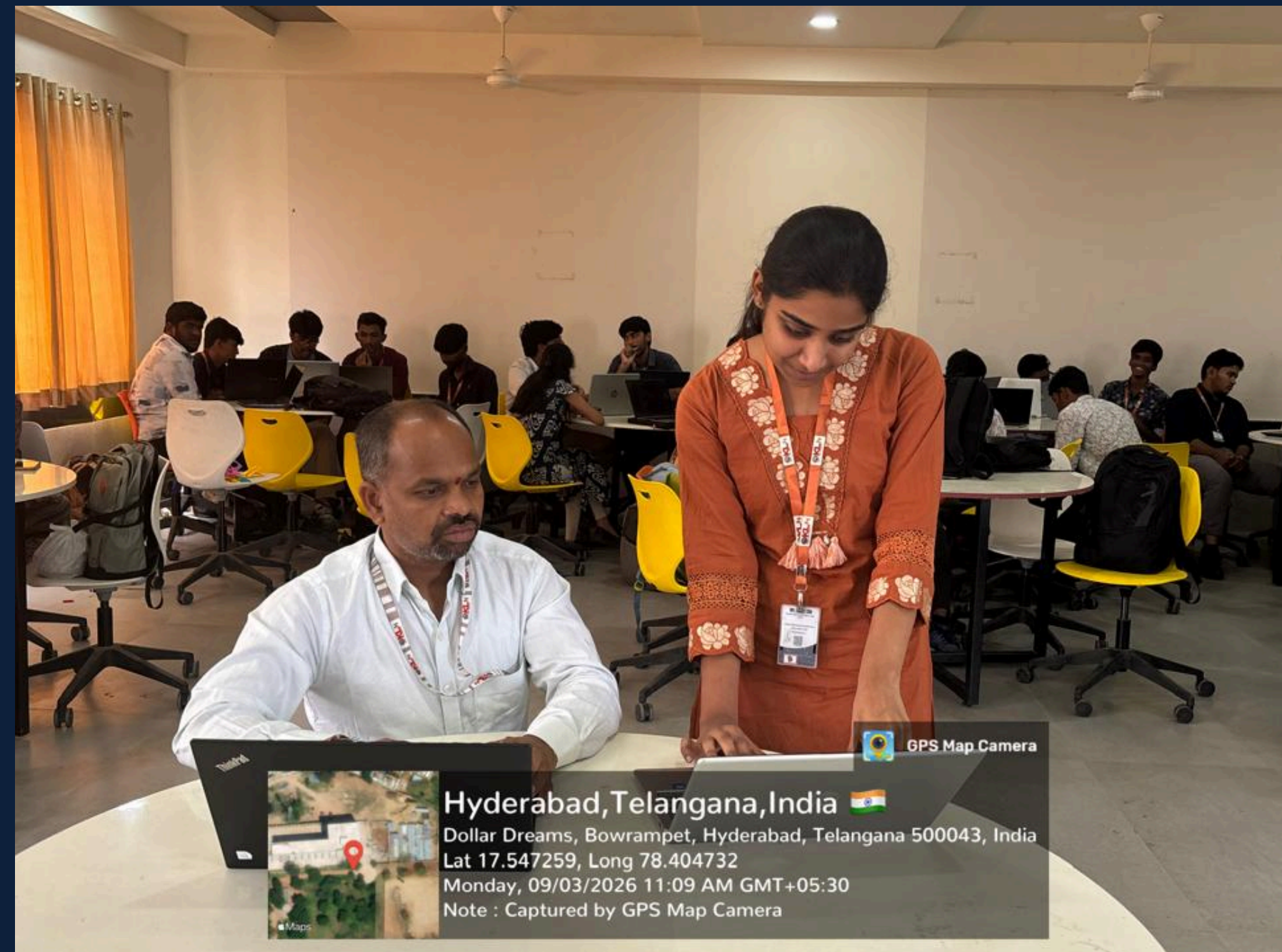
Runs directly in VS Code

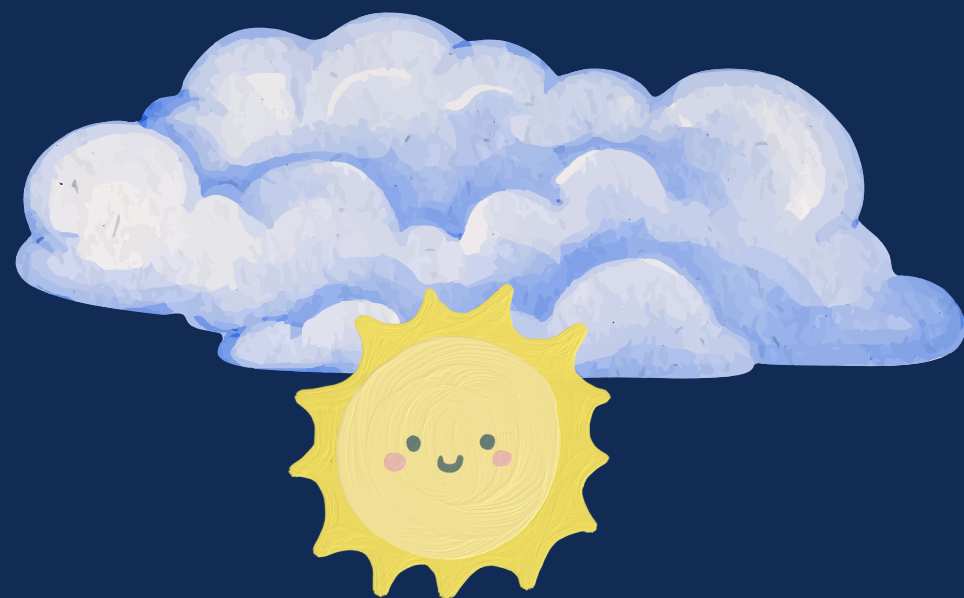
100%

Offline

No internet needed

GEO-TAG PICTURE





THANK YOU

